

Conversion Between Binary, Hexadecimal, and Octal

Conversion between binary and hexadecimal can be done by inspection since each hexadecimal digit corresponds to exactly four binary digits (bits).

$$10011101.010111_2 = 4D.5C_{16}$$

Conversion between binary to octal, except each octal digit corresponds to three binary digits, instead of four.

$$100101101011010_2 = 45,532_8$$